

WPT

What is Internet

The Internet is a global system that connects computers and devices worldwide, enabling communication, information sharing, and access to digital services.

- Connects people and devices globally
- Enables communication, learning, and business
- Supports digital services and cloud platforms

History of the Internet

- The Internet originated in the 1960s with the development of [ARPANET \(Advanced Research Projects Agency Network\)](#). It was designed to allow multiple computers to communicate within a single network an achievement that was revolutionary at the time.
- ARPANET used packet switching, a method that breaks data into packets for efficient transmission. In October 1969, the first message was successfully sent between two computers using ARPANET. From that moment, internet technology continued to evolve rapidly.

How Does the Internet Work

The internet is a global network of interconnected computers and devices that share information, resources, and services. It enables instant data exchange worldwide and powers everyday activities like emailing, browsing, streaming, and online gaming.

- The Internet connects billion users and billion websites using standard protocols like TCP/IP.
- Data is split into packets that travel through routers and switches.
- Every device has a unique IP address

Components of Internet

- **Packets**

Data sent over the Internet is split into small pieces called packets. Each packet has part of the data and information about where it's going. Packets travel through devices like routers and switches and are reassembled at the destination. This method, called packet switching,

- **Protocols**

- Protocols are rules that tell how data is sent, received, and understood over the Internet. They make sure packets arrive correctly and in order. Common protocols include:
 - **TCP/IP** – handles packet addressing and routing
 - **HTTP/HTTPS** – for web browsing
 - **FTP** – for file transfers
 - **SMTP/IMAP/POP3** – for email communication

Internet Set Up

- The Internet relies on physical infrastructure such as:
 - Optical fiber cables
 - Copper wires
 - Networking systems like LAN, WAN, and MAN
- Mobile technologies like 4G, 5G and Wi-Fi also depend on these physical cable networks to function.

How Does the Internet Work?

- The Internet operates using a client–server model:

Client: A device such as a laptop or smartphone connected to the Internet

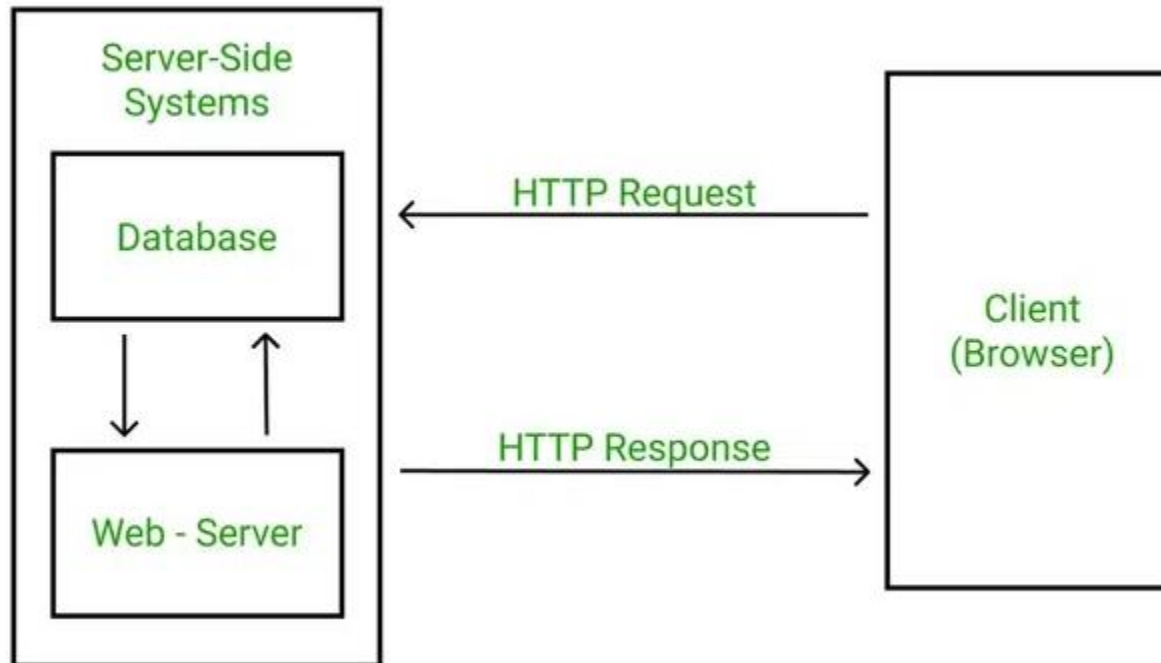
Server: Powerful computers that store websites and data

- **Step-by-step working:**
- The client sends a request through a browser
- The server identifies the website using its IP address
- DNS converts the domain name into an IP address
- The request is routed to the correct server
- The server processes the request and returns the webpage

- <https://www.ietpune.com/>
- WWW (World Wide Web)
- The World Wide Web is a collection of web pages and documents accessed through URLs (Uniform Resource Locators) on the Internet.
- It serves as an information retrieval system where documents are linked using hypertext or hypermedia. Hyperlinks allow users to click on words or phrases to easily access related information.

World Wide Web	Internet
All the web pages and web documents are stored there on the World wide web and to find all that stuff you will have a specific URL for each website.	The Internet is a global network of computers that is accessed by the World wide web.
The world wide web is a service.	The Internet is an infrastructure.
The world wide web is a subset of the Internet.	The Internet is the superset of the world wide web.
The world wide web is software-oriented.	The Internet is hardware-oriented.
The world wide web uses HTTP .	The Internet uses IP Addresses .
The world wide web can be considered as a book from the different topics inside a Library.	The Internet can be considered a Library.
Examples: Websites, e-commerce, blogs	Example: The network connecting all online services

Web Architecture



- **HTTP** stands for **Hypertext Transfer Protocol**, and it's the system that allows communication between web browsers (like Google Chrome or Firefox) and websites.
- When you visit a website, your browser uses HTTP to send a request to the server hosting that site, and the server sends back the data needed to display the page.
- Think of HTTP as a set of rules or a language used by your browser and the web server to talk to each other, ensuring that websites load properly when you type in their URLs.

HTTP : Step-by-Step Process

- 1) **Open Web Browser:** First, you open your web browser and type a website URL (e.g., www.example.com).
- 2) **DNS Lookup:** Your browser asks a [Domain Name System \(DNS\)](#) server to find out the IP address associated with that URL. (looking up the phone number of the website.)
- 3) **Send HTTP Request:** Once the browser has the website's IP address, it sends an HTTP **request** to the server. The request asks the server for the resources needed to display the page (like text, images, and videos).
- 4) **Server Response:** The server processes your request and sends back an HTTP **response**. This response contains the requested resources (like HTML, CSS, JavaScript) needed to load the page.
- 5) **Rendering the Web Page:** Your browser receives the data from the server and displays the webpage on your screen.

What is HyperText?

- **HyperText** is a way of structuring text so that it can contain links (called "hyperlinks") to other documents or resources. When you click on a link in a webpage, you are typically directed to another page or resource on the internet. HTML (HyperText Markup Language) is used to create and format this type of text for web pages.
- HTTP is the protocol used to transfer this hypertext between the web browser and the server, allowing you to click links and move around the web.

HTTP Request and Response

- An [HTTP request](#) is how your browser asks the server for something. It includes:
- **HTTP Version:** The version of HTTP (like HTTP/1.1 or HTTP/2) being used.
- **URL:** The specific address of the resource (e.g., `https://www.example.com/about`).
- **HTTP Method:** The type of action being requested (e.g., GET to retrieve information or POST to send data).
- **HTTP Request Headers:** Extra information about the request, like what kind of browser you're using or what kind of content you're expecting.
- **HTTP Request Body:** In some cases, the request will include a body that contains data (e.g., when you submit a form).

HTTP Response

- An [HTTP response](#) is the server's answer to your request. It includes:
- **HTTP Status Code:** A number that tells you if the request was successful or not (e.g., 200 OK means everything is fine, 404 Not Found means the requested page doesn't exist).
- **Response Headers:** Information about the response, like what kind of data is being sent (e.g., Content-Type: text/html means it's an HTML page).
- **Response Body:** The content that the server sends back (e.g., HTML code that the browser will use to display the webpage).

HTTP Status Code

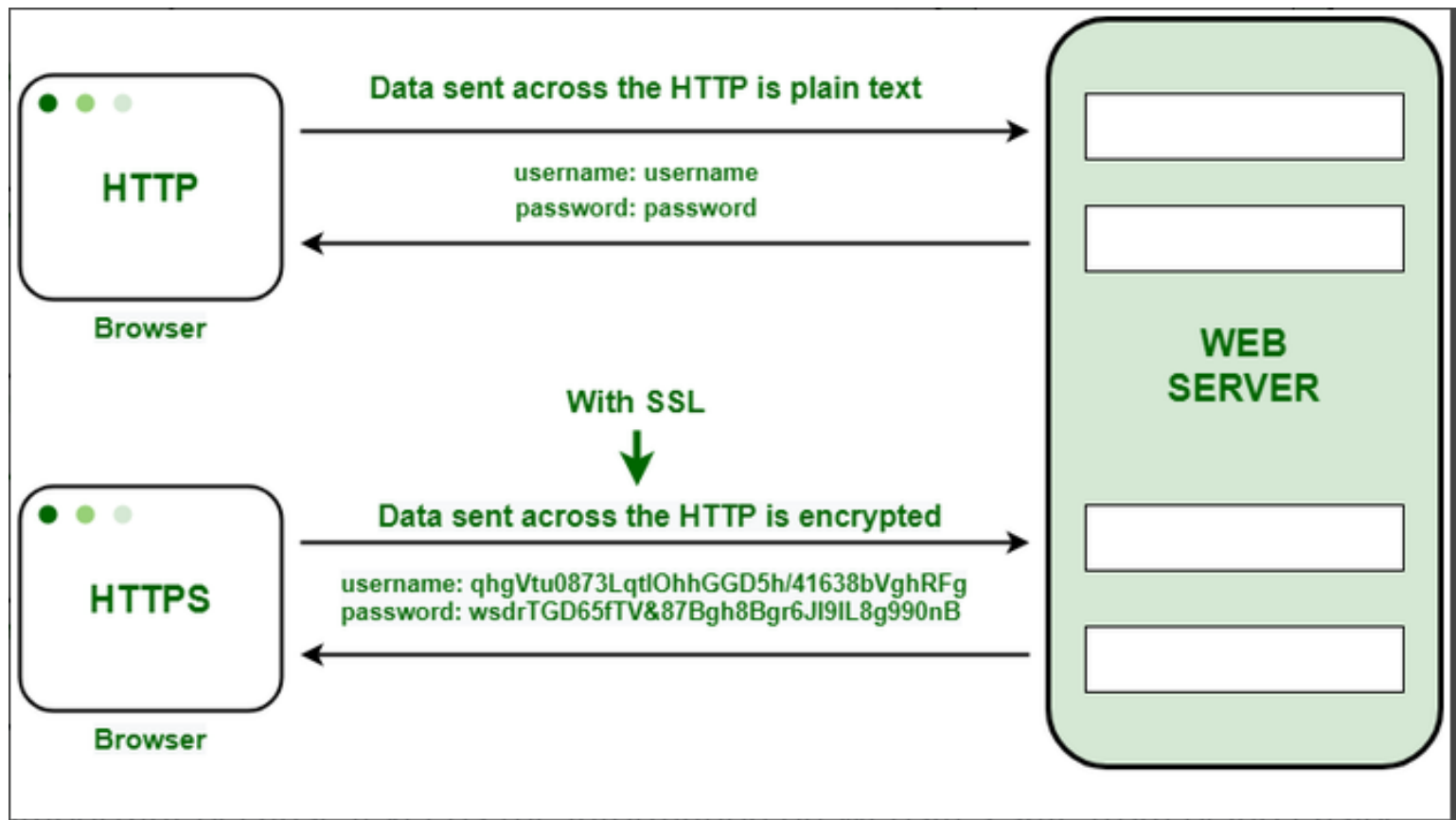
- **HTTP Status codes** are three-digit numbers that servers use to tell your browser what happened with the request you sent.
- 1) Informational (**1xx**): These codes just give you information
 - 2) Successful (**2xx**): These codes tell you everything went fine (e.g., 200 OK means the request was successful).
 - 3) Redirection (**3xx**): These codes tell the browser to take additional action (e.g., 301 Moved Permanently means the requested page has moved to a new address).
 - 4) Client Error (**4xx**): These codes indicate that there was a problem with your request (e.g., 404 Not Found means the page doesn't exist).
 - 5) Server Error (**5xx**): These codes tell you that something went wrong on the server side (e.g., 500 Internal Server Error means the server had an issue processing the request).

Internet Protocol

- HTTP stands for **Hypertext Transfer Protocol**, and it's the system that allows communication between web Client (Browser) and Web Server
- Communication means Data Exchange

So if you want Data Security we can use HTTPS protocol

- HTTPS establishes the communication between the browser and the web server. It uses the **Secure Socket Layer (SSL)** and **Transport Layer Security (TLS)** protocol for establishing communication.
- HTTPS uses the conventional HTTP protocol and adds a layer of SSL/TLS over it. The workflow of HTTP and HTTPS remains the same
- The SSL connection is responsible for the encryption and decryption of the data that is being exchanged to ensure data safety.



HTTP	HTTPS
HTTP stands for HyperText Transfer Protocol.	HTTPS stands for HyperText Transfer Protocol Secure.
URL begins with “http://”.	URL starts with “https://”.
HTTP Works at the Application Layer.	HTTPS works at Transport Layer.
HTTP speed is faster than HTTPS.	HTTPS speed is slower than HTTP.

What is HTML

- HTML stands for **Hyper Text Markup Language**
- HTML is the standard markup language for creating static Web pages
- HTML describes the structure of a Web page which contains elements
- HTML elements tell the browser how to display the content

```
<!DOCTYPE html>
<html>
<head>
<title>Page Title</title>
</head>
<body>

|

</body>
</html>
```

- HTML Documents Contains an Elements
- Element Contains <Start Tag> Content </End Tag>
- Some Elements Are Empty
